

● EASY

● ALGEBRA

Linear functions

06

10 questions · ~15 min

Q1

ALGEBRA

In science class, Diego conducted an experiment to learn about evaporation. Diego measured the height of fluid in a beaker over a period of time. The function $f(x) = 39 - 0.18x$ gives the estimated height, in centimeters cm, of the fluid in the beaker x days after the start of the experiment. Which of the following is the best interpretation of 39 in this context?

- A. The estimated height, in cm, of the fluid at the start of the experiment
- B. The estimated height, in cm, of the fluid at the end of the experiment
- C. The estimated change in the height, in cm, of the fluid each day
- D. The estimated number of days for all the fluid to evaporate

$$gx = 11x + 4$$

For the given linear function g , which table shows three values of x and their corresponding values of gx ?

A.

x	gx
-1	7
0	11
1	15

B.

x	gx
-1	-4
0	0
1	4

C.

x	gx
-1	-7
0	4
1	15

D.

x	gx
-1	-11
0	0
1	11

Q3

ALGEBRA

The function g is defined as $g(x) = 5x + a$, where a is a constant. If $g(4) = 31$, what is the value of a ?

- A. 30
- B. 22
- C. 11
- D. -23

Q4

ALGEBRA

The function f is defined by $fx = -3x + 60$. What is the value of fx when $x = -8$?

- A. 49
- B. 52
- C. 57
- D. 84

Q5

ALGEBRA

For a training program, Juan rides his bike at an average rate of 5.7 minutes per mile. Which function m models the number of minutes it will take Juan to ride x miles at this rate?

- A. $mx = \frac{x}{5.7}$
- B. $mx = x + 5.7$
- C. $mx = x - 5.7$
- D. $mx = 5.7x$

Q6

GRID-IN ALGEBRA

If $y = 5x + 10$, what is the value of y when $x = 8$?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

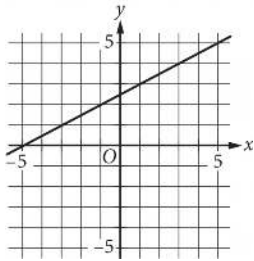
Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

The function g is defined by $gx = 10x + 8$. What is the value of gx when $x = 8$?

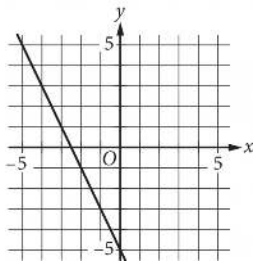
- A. 0
- B. 8
- C. 10
- D. 88

Which of the following is the graph of the equation $y = 2x - 5$ in the xy -plane?

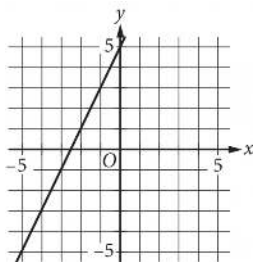
A.



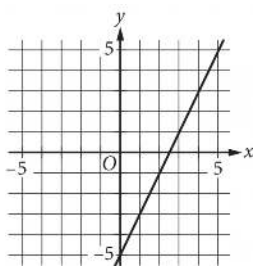
B.



C.



D.



Q9

ALGEBRA

x	$f(x)$
0	29
1	32
2	35

For the linear function f , the table shows three values of x and their corresponding values of $f(x)$. Which equation defines $f(x)$?

- A. $f(x) = 3x + 29$
- B. $f(x) = 29x + 32$
- C. $f(x) = 35x + 29$
- D. $f(x) = 32x + 35$

The function f is defined by $fx = \frac{1}{10}x - 2$. What is the y -intercept of the graph of $y = fx$ in the xy -plane?

A. $-2, 0$

B. $0, -2$

C. $0, \frac{1}{10}$

D. $\frac{1}{10}, 0$